

Q: List elements of quality plan. Some system analysts claim that requirements that belong to certain software quality factors should not be considered when preparing a project development plan. Do you agree with this claim? If yes – list the software quality factors that should not be considered. If not – explain your arguments.

(a) Elements of a Quality Plan

The elements of a Quality Plan are:

List of Quality Goals

Defines measurable quality requirements (e.g., availability, reliability, response time).

Review Activities

Planned design reviews, design inspections, and code inspections.

Specifies scope, type, schedule, procedures, and responsible persons.

Planned Software Tests

Unit tests, integration tests, and system tests.

Includes test types, schedule, procedures, and responsibilities.

Acceptance Tests for Externally Developed Software

Acceptance tests for purchased software, subcontractor software, and customer-supplied software.

Configuration Management Tools and Procedures

Defines version control and change control:

Who makes changes, when, why, and what is changed.

Quality Plan Document, Format, and Approval

The quality plan may be part of the development plan or a separate document and must be reviewed and approved.

(b) Do you agree with the claim?

No, I do not agree with this claim.

(c) Arguments (Why the claim is not correct)

Software quality factors affect system architecture

Quality factors such as performance, security, reliability, and usability influence system design.

Ignoring them during planning may cause major redesign later.

Quality factors affect cost and schedule

If quality requirements are ignored in planning, adding them later increases cost and causes delays.

Different projects have different priorities, but no factor should be ignored

Some quality factors are more important for certain systems, but none should be excluded completely.

For example, security is critical for financial systems, performance is critical for real-time systems.

Quality standards require early planning of quality

International quality models such as ISO 25010 define software quality characteristics that must be addressed from the planning stage.

Project planning practices recommended by IEEE also integrate quality requirements into development planning.